

Turbocharger Installation Manual

汽车涡轮增压器的按照说明书 1.0.

1. Pre-Installation Preparation & Inspection

1.1 Tools & Materials Required

- Socket wrench set, torque wrench, flathead/Phillips screwdrivers
- New turbocharger (verify part number matches engine)
- **New gaskets, O-rings, crush washers** (never reuse old ones)
- Clean engine oil (per manufacturer spec), oil filter, air filter
- Anti-seize compound (high-temperature)
- Clean rags, solvent, compressed air

1.2 Turbo & Engine Pre-Check

- Remove plastic/foam blanking plugs from turbo oil/coolant portsGarrett Motion
- Inspect compressor/turbine wheels: no damage, spin freely by hand
- Clean engine manifold/block mounting surfaces: remove oil, sludge, old gasket residue
- Inspect oil feed/drain lines: replace kinked/damaged lines; use new oil inlet line (recommended)Garrett Motion
- Check exhaust manifold for warping/cracks; inspect intercooler/intake hoses for leaks/blockages
- Drain engine oil/coolant if required

2. Installation Steps

2.1 Mount the Turbocharger

1. Place **new gasket/O-ring** between turbo and exhaust manifold/engine blockGarrett Motion
2. Align turbo precisely; hand-thread mounting bolts first
3. Tighten bolts/nuts to **manufacturer torque specs** (do not over-tighten)Garrett Motion
4. Reconnect exhaust pipe; use V-band clamps (align and torque correctly)

2.2 Oil System Installation (Critical)

1. Install oil drain line first; ensure drain port tilt $\leq 20^\circ$ from bottom center
2. Fill turbo oil inlet port with **clean engine oil** until overflowing
3. Connect oil feed line; use new gaskets/crush washers; avoid silicone on oil flanges
4. Spin compressor wheel by hand 5–10 times to distribute oilGarrett Motion
5. Install new oil filter; pre-fill with clean oil if accessible

2.3 Air & Coolant System Installation

1. Connect intake/outlet air hoses to compressor housing; tighten clamps for airtight sealGarrett Motion
2. Install intercooler piping (if equipped); clean intercooler with solvent/air before connection

3. Reconnect coolant lines (if applicable); use new O-rings/gaskets
4. Check all connections for clearance; no contact with hot/exposed components

3. Post-Installation Priming & Testing

3.1 Oil Priming (No Engine Start)

- Disable ignition/fuel system; crank engine for **10–15 seconds** to prime oil flow^{Garrett Motion}
- Repeat 2–3 times until oil pressure builds; check for oil leaks at turbo connections

3.2 Engine Start & Leak Inspection

1. Start engine; let idle for **3–4 minutes**^{Garrett Motion}
2. Inspect all joints for oil, exhaust, or air leaks; fix immediately if found
3. Check oil pressure ($\geq 0.5 \text{ kgf/cm}^2$ / 7 psi within 3 seconds of start)
4. For VNT turbos: verify actuator operation is normal^{Garrett Motion}

3.3 Final Checks

- Stop engine; recheck oil level and top up if needed
- Inspect all fasteners for torque; ensure no loose clamps/bolts
- Replace air filter; ensure intake system is free of debris

4. Safety & Warranty Notes

- Do not modify turbo settings/calibration; avoid adjusting VNT vane positions
- Do not use liquid gasket/sealant on oil ports (risk of blockage)
- Follow engine manufacturer's oil change intervals; use high-quality oil
- Keep turbo away from heat sources; avoid bending oil/coolant lines